

Regular Graphs

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July 31, 2026

Definition 0.1 (k-Regular Graph). A graph $G = (V, E)$ is called **k -regular** if every vertex has degree exactly k neighbours. Formally,

$$\deg(v) = k, \quad \forall v \in V(G).$$

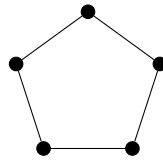
That is, every vertex is incident to exactly k edges.

Example 0.2 (1-Regular Graph). Every vertex has degree 1. Such graphs consist of disjoint edges.



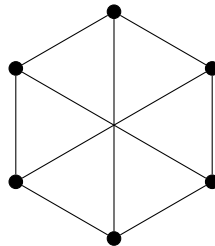
Each vertex is incident to exactly one edge.

Example 0.3 (2-Regular Graph). Every vertex has degree 2. A connected 2-regular graph is a cycle.



Each vertex has two neighbours.

Example 0.4 (3-Regular (Cubic) Graph). A graph is **3-regular** if every vertex has degree 3.



Every vertex is incident to exactly three edges.

Definition 0.5 ($2k$ -Regular Graph). A graph G is said to be **$2k$ -regular** if every vertex has even degree $2k$, i.e.,

$$\deg(v) = 2k, \quad \forall v \in V(G).$$

Examples include

k	Graph Type
1	2-regular
2	4-regular
3	6-regular
4	8-regular

Remark 0.6. The paper focuses on $2k$ -regular graphs because every vertex has even degree. This enables the application of *Petersen's 2-Factor Theorem*, which states that every $2k$ -regular graph can be decomposed into k edge-disjoint 2-factors. This decomposition is fundamental to the proofs developed later in the paper.